

# Ray theory

# From last lecture

Plane wave ansatz

$$u_i(x, t) = f(t - p_j x_j) a_i$$

Christoffel equation for plane waves:

$$\det[\Gamma(\mathbf{p}) - \mathbf{I}] = 0$$

$$\Gamma_{ik}(\mathbf{p}) = \frac{1}{\rho} A_{ijkl} p_j p_l$$

Definition of the slowness surface:

$$\det[\Gamma(\mathbf{p}) - \mathbf{I}] = 0$$



$$\begin{aligned}\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} &= \mathbf{0}, & \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{H} + \frac{\partial \mathbf{D}}{\partial t} &= \mathbf{J}, & \nabla \cdot \mathbf{D} &= \rho,\end{aligned}$$

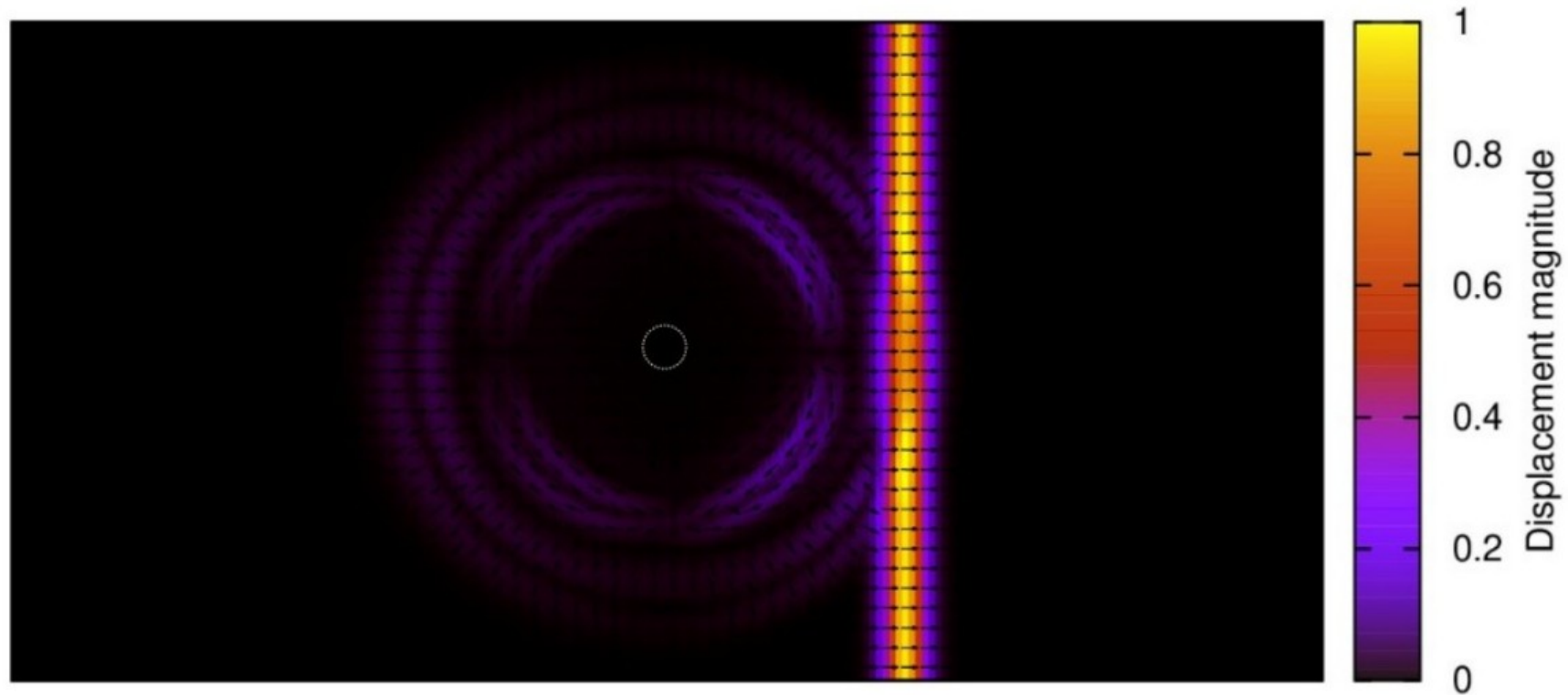
Quantum mechanics

$$i\hbar \frac{\partial \Psi}{\partial t} = H\Psi$$

Classical mechanics

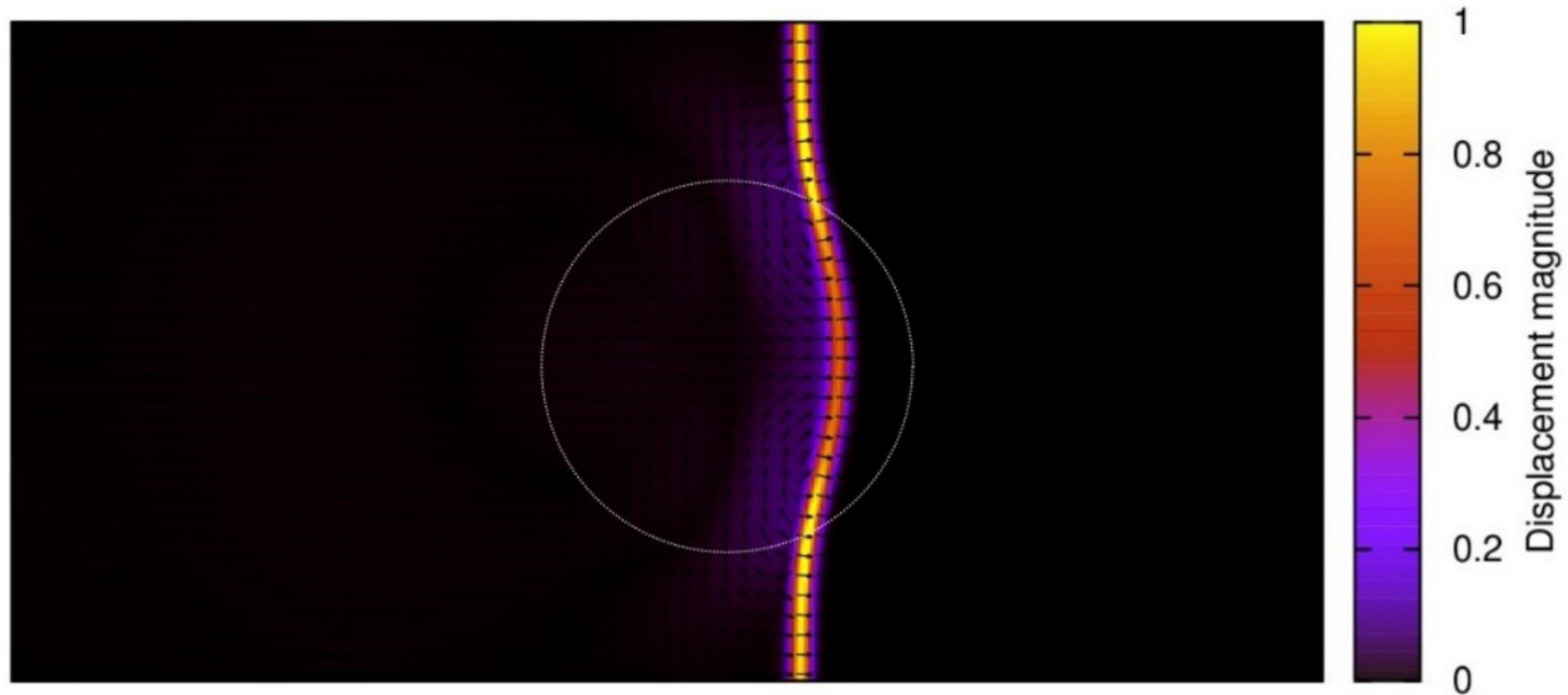
$$\begin{aligned}\frac{d\mathbf{q}}{dt} &= \frac{\partial H}{\partial \mathbf{p}} \\ \frac{d\mathbf{p}}{dt} &= -\frac{\partial H}{\partial \mathbf{q}}\end{aligned}$$

Length scales of heterogeneity



Scattering regime:

$$\frac{L}{\lambda} \approx 1$$



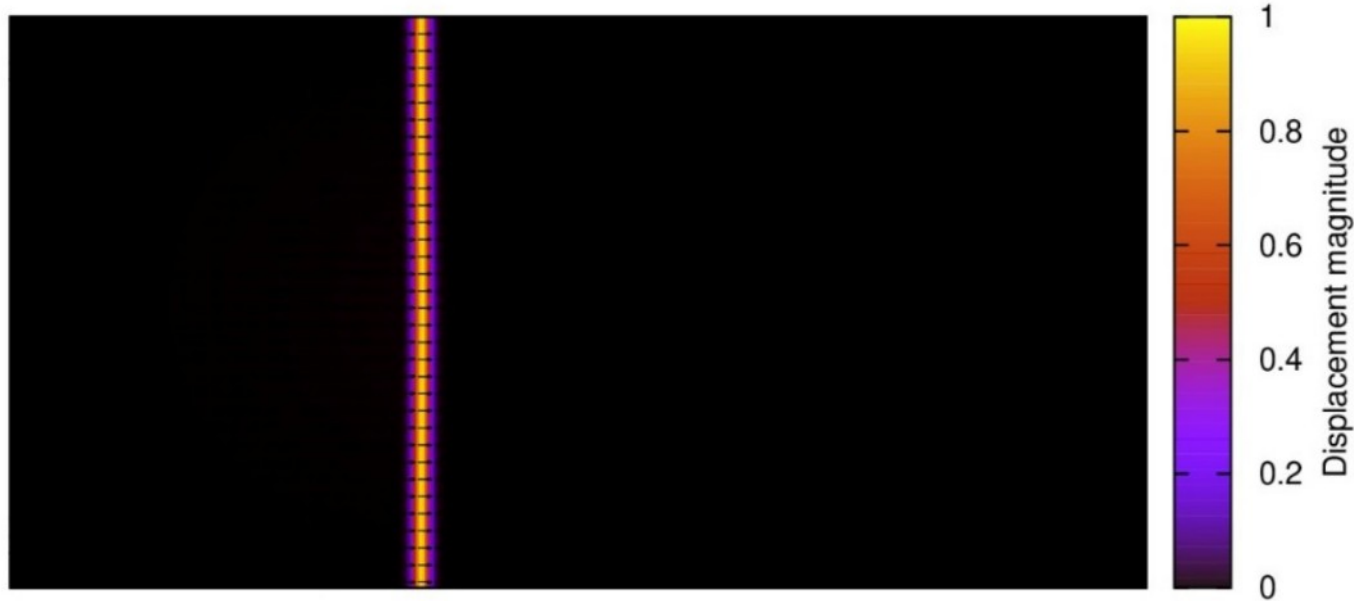
Geometric optics regime:

$$\frac{L}{\lambda} \gg 1$$

Ray series ansatz



# Canonical problem

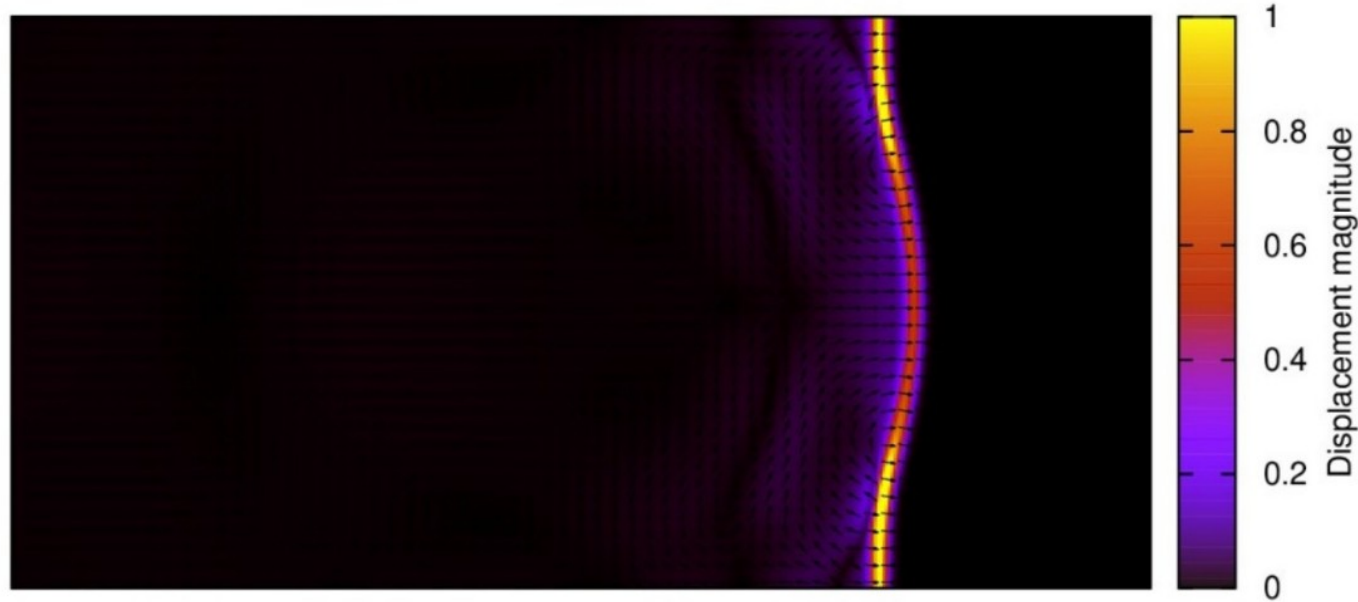


Start with a plane wave within the homogeneous left half plane:

$$u_i(x, t) = f(t - p_i^0 x_i) a_i^0$$

Can we extend this solution into the heterogeneous right half plane?

# Initial ansatz



As an initial guess, we look for a solution of the form:

$$u_i(x, t) = f[t - T(x)]a_i(x)$$

# Travel time and local slowness

Assumed solution takes the form:

$$u_i(x, t) = f[t - T(x)]a_i(x)$$

Introduction of the **travel time** function allows for curved wavefronts:

$$T(x) = \text{constant.}$$

The **local slowness vector** is orthogonal to the wavefronts:

$$p_i = \frac{\partial T}{\partial x_i}$$

Polarisation vector allowed to vary with position – accounts for **focusing effects** and for variations in anisotropic properties with position

# The ray series

Harmonic decomposition solution:

$$u_i(x, t) = \frac{1}{2\pi} \int_{\mathbb{R}} \tilde{f}(\omega) e^{i\omega[t-T(x)]} a_i(x) \, d\omega,$$

This is a convolution of  $f$  with the singular solution:

$$u_i(x, t) = \delta[t - T(x)] a_i(x)$$

Experience has shown that we should generalise the singular part to:

$$u_i(x, t) = \frac{1}{2\pi} \int_{\mathbb{R}} \sum_{n=0}^{\infty} \frac{a_i^n(x)}{(i\omega)^n} e^{i\omega[t-T(x)]} \, d\omega,$$

First term ( $n=0$ ) in the **ray series** is the singular one, while successive ones become smoother and smoother

# The eikonal equation

# Frequency-domain equations

Linearised equations with no force term:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left( A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = 0,$$

Transforming to the frequency-domain:

$$-\omega^2 \rho u_i - \frac{\partial}{\partial x_j} \left( A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = 0$$

Ray series in the frequency-domain:

$$u_i(x, \omega) = \sum_{n=0}^{\infty} \frac{a_i^n(x)}{(i\omega)^n} e^{-i\omega T(x)}.$$

We can now substitute in our ansatz and see what happens!

# Lots of differentiation...

Example step from the calculation:

$$\frac{\partial u_k}{\partial x_l} = \sum_{n=0}^{\infty} \frac{1}{(i\omega)^n} \left( \frac{\partial a_k^n}{\partial x_l} - i\omega p_l a_k^n \right) e^{-i\omega T},$$

Factor of frequency is added each time we differentiate the phase term

# Result of the calculation

Equations of motion imply:

$$\sum_{n=0}^{\infty} \frac{1}{(i\omega)^n} \left[ \frac{\partial}{\partial x_j} \left( A_{ijkl} \frac{\partial a_k^n}{\partial x_l} \right) - i\omega \frac{\partial}{\partial x_j} (A_{ijkl} p_l a_k^n) \right. \\ \left. - i\omega A_{ijkl} p_j \frac{\partial a_k^n}{\partial x_l} + (i\omega)^2 A_{ijkl} p_j p_l a_k^n - (i\omega)^2 \rho a_i^n \right] e^{-i\omega T} = 0.$$

Common phase term cancels

These equations must hold for **all frequencies** – this implies that the **coefficients of each power of frequency must separately vanish**

Compare with power series solution of ODEs



# Results of the calculation

From the highest order coefficient, we have:

$$[A_{ijkl}p_jp_l - \rho\delta_{ik}]a_k^0 = 0,$$

At the next order:

$$[A_{ijkl}p_jp_l - \rho\delta_{ik}]a_k^1 = \frac{\partial}{\partial x_j}(A_{ijkl}p_la_k^0) + A_{ijkl}p_j\frac{\partial a_k^0}{\partial x_l},$$

For the general term with  $n > 1$ :

$$[A_{ijkl}p_jp_l - \rho\delta_{ik}]a_k^n = \frac{\partial}{\partial x_j}(A_{ijkl}p_la_k^{n-1}) + A_{ijkl}p_j\frac{\partial a_k^{n-1}}{\partial x_l} - \frac{\partial}{\partial x_j}(A_{ijkl}\frac{\partial a_k^{n-2}}{\partial x_l})$$

First equation determines **p**, while the later ones successively determine the polarisation vectors at different orders

Where have we made an approximation?

# A short cut

In this course we will consider only the equation derived from the **highest-order** term in the expansion

If that is all that we required, we can systematically drop all but the highest order terms within the previous derivation:

$$\frac{\partial u_k}{\partial x_l} = \sum_{n=0}^{\infty} \frac{1}{(i\omega)^n} \left( \frac{\partial a_k^n}{\partial x_l} - i\omega p_l a_k^n \right) e^{-i\omega T} \approx -i\omega p_l a_k^0 e^{-i\omega T}$$

This saves a lot of time!

You can practice these steps in Example sheet 3

# Highest order condition

From the highest order coefficient, we have:

$$[A_{ijkl}p_jp_l - \rho\delta_{ik}]a_k^0 = 0,$$

This motivates the definition of a local Christoffel matrix:

$$\Gamma_{ik}(\mathbf{x}, \mathbf{p}) = \frac{1}{\rho(\mathbf{x})} A_{ijkl}(\mathbf{x}) p_j p_l.$$

And so write the above condition as:

$$\{\Gamma[\mathbf{x}, \mathbf{p}(\mathbf{x})] - \mathbf{I}\} \mathbf{a}^0(\mathbf{x}) = \mathbf{0},$$

Locally, things look identical to plane waves!

# The eikonal equation

The equation

$$\{\Gamma[\mathbf{x}, \mathbf{p}(\mathbf{x})] - \mathbf{I}\} \mathbf{a}^0(\mathbf{x}) = \mathbf{0},$$

has non-trivial solutions for the zeroth-order polarisation vector if and only if:

$$\det\{\Gamma[\mathbf{x}, \mathbf{p}(\mathbf{x})] - \mathbf{I}\} = 0,$$

This is a non-linear first order PDE for the travel time called the **eikonal equation**

If the local slowness surfaces are everywhere non-degenerate:

$$||\mathbf{p}(\mathbf{x})||^2 c_k[\mathbf{x}, \hat{\mathbf{p}}(\mathbf{x})]^2 - 1 = 0$$

The method of characteristic

# Ray Hamiltonian

Recall the eikonal equation is:

$$||\mathbf{p}(\mathbf{x})||^2 c_k[\mathbf{x}, \hat{\mathbf{p}}(\mathbf{x})]^2 - 1 = 0$$

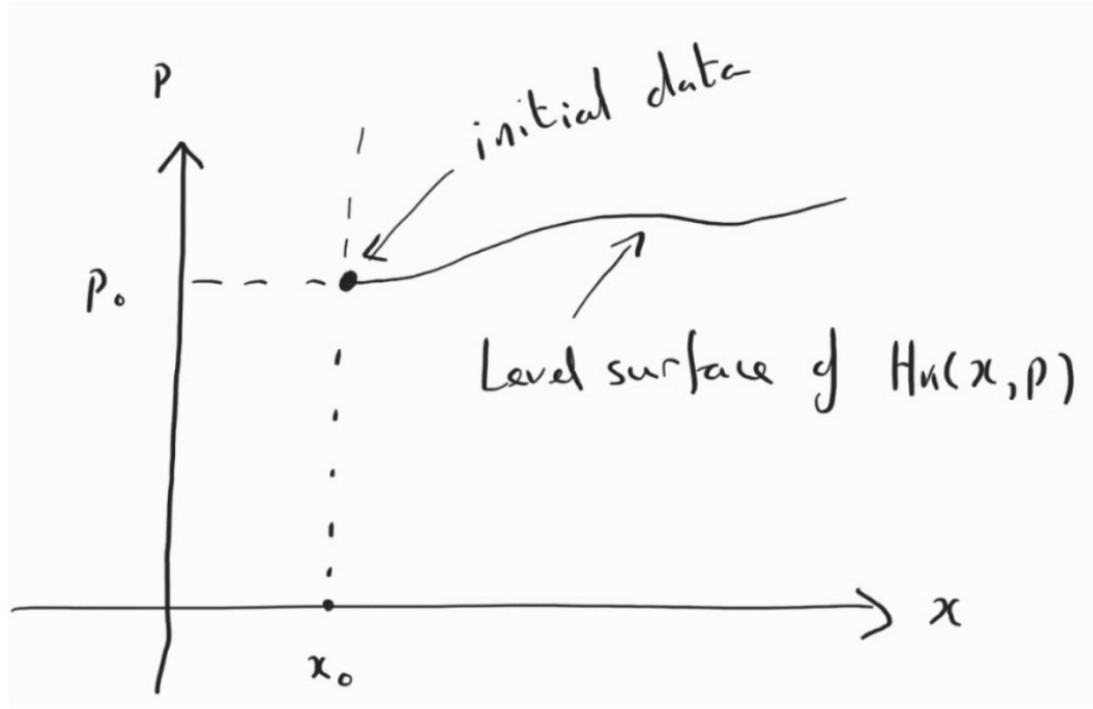
If we defined the **ray Hamiltonian** for the  $k$ th sheet of the slowness surface:

$$H_k(\mathbf{x}, \mathbf{p}) = \frac{1}{2} ||\mathbf{p}(\mathbf{x})||^2 c_k[\mathbf{x}, \hat{\mathbf{p}}(\mathbf{x})]^2,$$

We can write the eikonal equation in the form:

$$H_k[\mathbf{x}, \mathbf{p}(\mathbf{x})] = \frac{1}{2}, \quad \mathbf{p}(\mathbf{x}) = (\nabla T)(\mathbf{x}).$$

# Geometric interpretation



Solution of eikonal equation is a 3D level surface of the Hamiltonian in a 6D phase space

Initial data specifies a 2D level surface in the phase space

The problem is how to extend this 2D surface into the appropriate 3D level surface

This is where Hamilton's equations enter into the method

# Hamilton's equations

Initial data extended through integration of Hamilton's equations:

$$\frac{dx_i}{d\sigma} = \frac{\partial H_k}{\partial p_i}, \quad \frac{dp_i}{d\sigma} = -\frac{\partial H_k}{\partial x_i}$$

$$x_i(0) = x_i^0, \quad p_i(0) = p_i^0.$$

This works because the Hamiltonian is constant along the integral curves:

$$\frac{d}{d\sigma} H[\mathbf{x}(\sigma), \mathbf{p}(\sigma)] = \frac{\partial H}{\partial x_i} \frac{dx_i}{d\sigma} + \frac{\partial H}{\partial p_i} \frac{dp_i}{d\sigma} = -\frac{dp_i}{d\sigma} \frac{dx_i}{d\sigma} + \frac{dx_i}{d\sigma} \frac{dp_i}{d\sigma} = 0,$$

There are a few subtler points to the method that are being ignored



# Recovering the travel time

**Rays** are the projection of integral curves into the **position-domain**

Rate of change of the travel time along a ray:

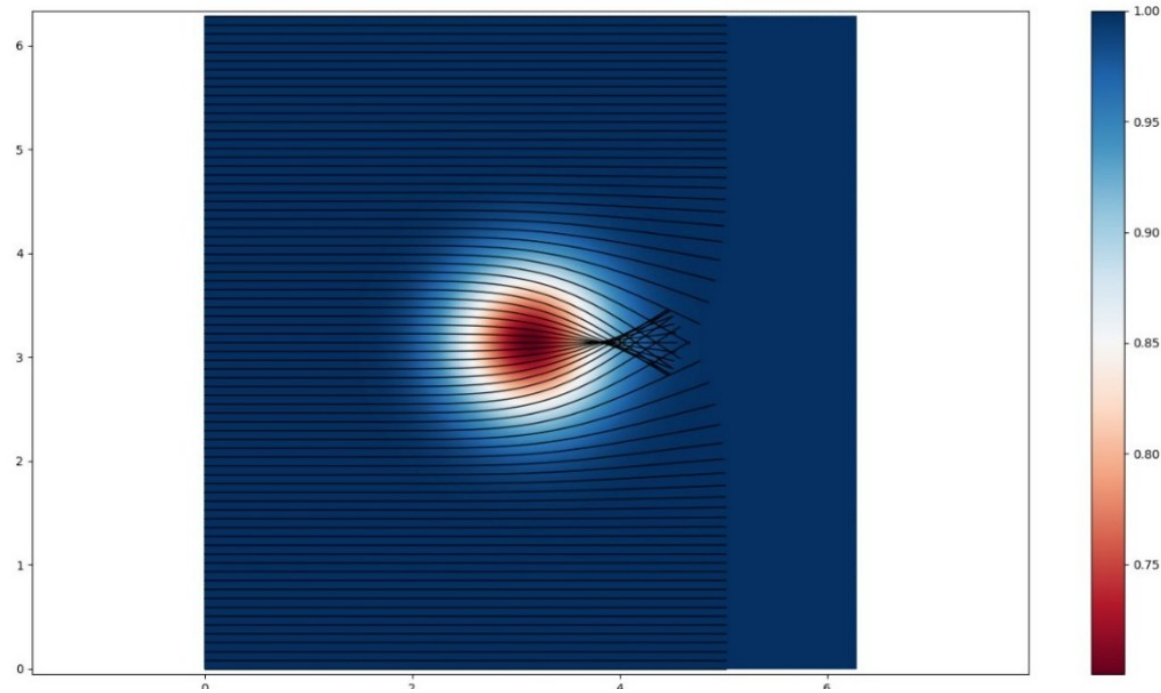
$$\frac{d}{d\sigma} T[\mathbf{x}(\sigma)] = \frac{\partial T}{\partial x_i} \frac{dx_i}{d\sigma} = p_i \frac{\partial H}{\partial p_i}$$

Right hand side simplified using Euler's homogeneous function theorem;

$$\frac{d}{d\sigma} T[\mathbf{x}(\sigma)] = 1.$$

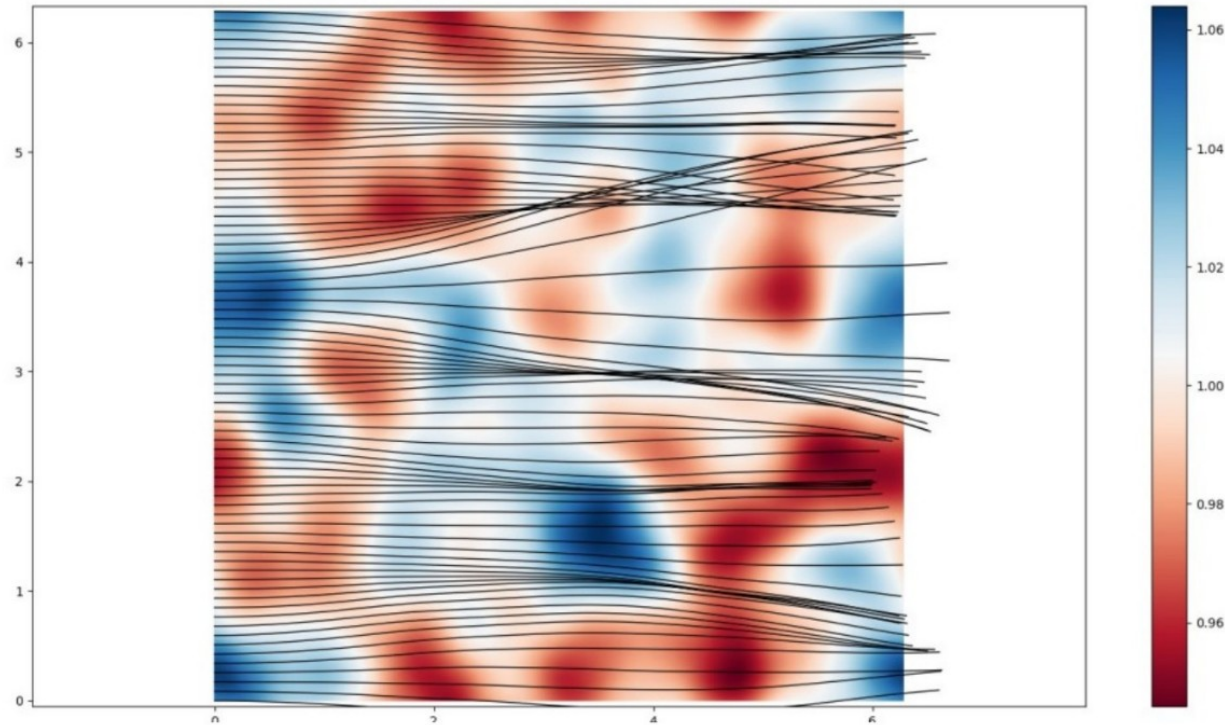
**The generating parameter along the ray is equal to the travel time**

# Caustics



If rays for two different integral curves touch, then the travel time need not be well-defined. These points are known as **caustics**. The **ray theoretic amplitude is infinite** at such points, while the true amplitude is large but finite

# Caustics



Within heterogeneous materials, caustics occur frequently – c.f. Penrose Hawking singularity theorems in general relativity